

Winter 2026

ENERGY295-Electrochemical Energy Storage Systems: Modeling and Estimation

Lecture 12

SIMONA ONORI, Stanford University

Energy Science and Engineering

Electrical Engineering (courtesy)

Precourt Institute of Energy



solving energy
for humanity.

Stanford
School of Sustainability
Energy Science & Engineering

Tomorrow 02/26

- Seminar at 4pm in 365 Green by Prof. Monica Marinescu, Imperial College London, UK

IMPERIAL

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DR
Monica Marinescu

BIO

I am interested in understanding how physical and chemical processes impact the behaviour of energy storage devices in applications. This involves identifying the mechanisms that limit behaviour under real conditions, which I do by building electrochemical and physical models. I use experimentally observed behaviour to parametrise and validate these models, and use them further to explore ways in which their performance can be improved. I also synthesise this knowledge into reduced order models, which can be used for control algorithms in a given application.

I am working on lithium ion batteries, lithium sulfur batteries, supercapacitors and Li-ion capacitors.

Editorial Boards

Guest Lecture: Batteries in the Real World - Industry Trends, Supply Chains, and Career Paths

Yen T. Yeh

*Executive Director,
Volta Foundation*



- Over a decade of experience in battery system development at MIT, Apple, and leading startups.
- Keen on promoting innovation and sustainability within the battery industry.
- Building strong partnerships to promote a vibrant battery ecosystem globally through the Volta Foundation

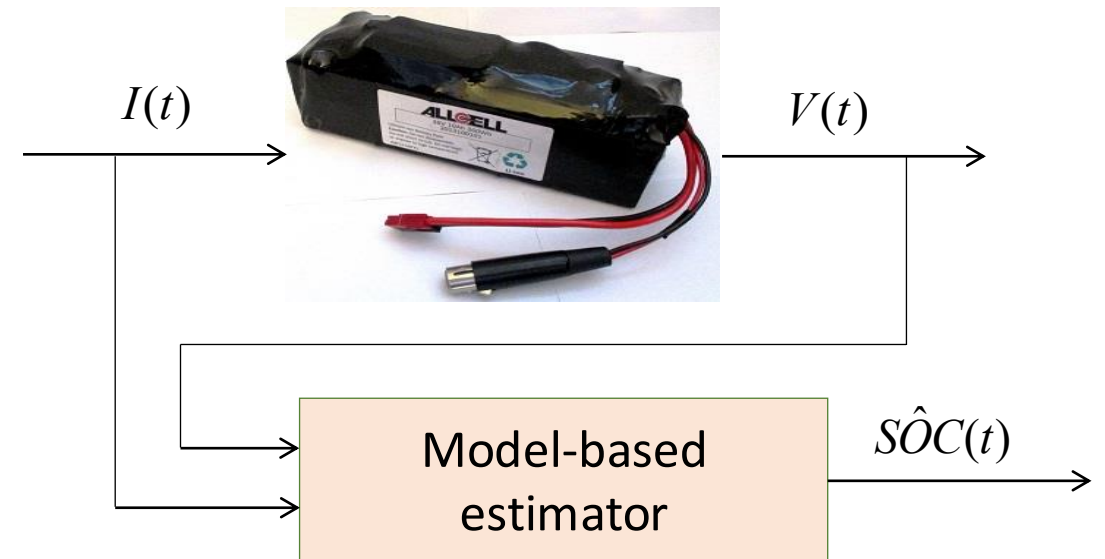
March 2nd, 10:30am

Hartley Conference Room



Model-based Observer

- Observer Design: Needed to determine the internal states of an system from a given set of input/output measurements
- We use feedback to correct the mathematical model-based estimator



Observer

- The design of a state observer consists of introducing an auxiliary dynamical system, called an estimator, which uses available measurements and inputs to generate an estimate of the system states.
- Hypothesis → the measured output and input are noise free (no bias, no uncertainty) → Luenberger Observer

Observability

The system is **observable** if:

The initial state $x(0)$ can be uniquely determined from knowledge of the input $u(t)$ and output $y(t)$ over a finite time interval.

$$y(t) = Ce^{At}x(0) + C \int_0^t e^{A(t-\tau)} u(\tau) B d\tau$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} u(\tau) B d\tau$$

→ If we know $x(0)$, then we know $x(t)$!

Observability

Given

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t)\end{aligned}$$

Observability matrix is:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

Observability test

System is observable $\Leftrightarrow \text{rank}(\mathcal{O}) = n$

Estimation Schemes

Assume the system model is of the form:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), & x(0) \text{ is unknown} \\ y(t) &= Cx(t)\end{aligned}$$

where

1. A , B , and C are known
2. $u(t)$ is known
3. Measurable output are $y(t)$, $C \neq I$ (identity matrix)

GOAL:

- Develop a dynamics system whose state

$$\hat{x}(t) = x(t) \text{ for all } t \geq 0$$

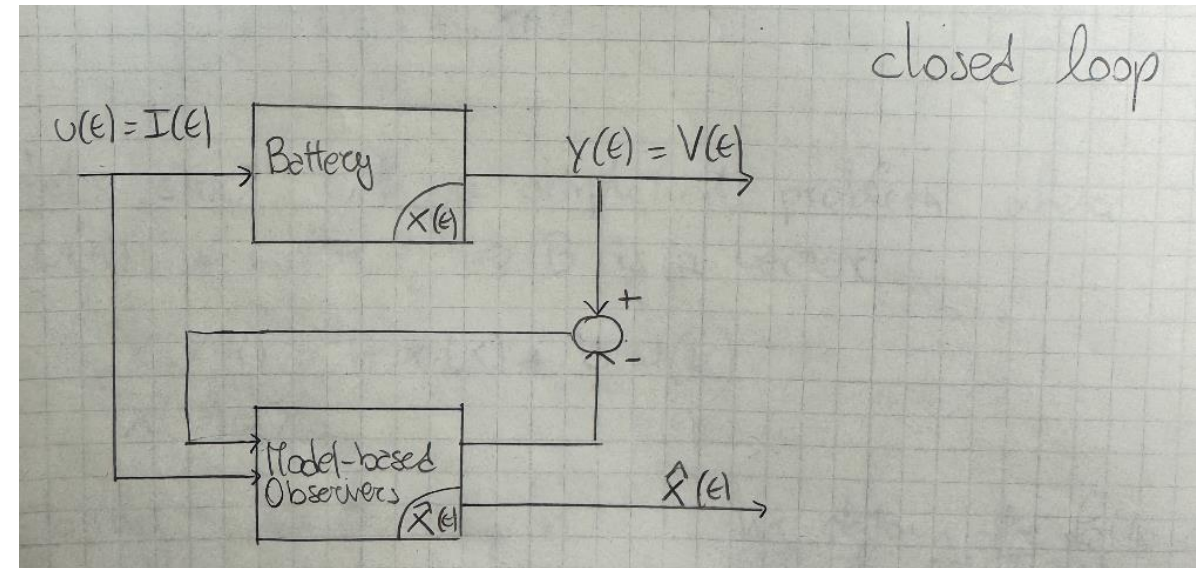
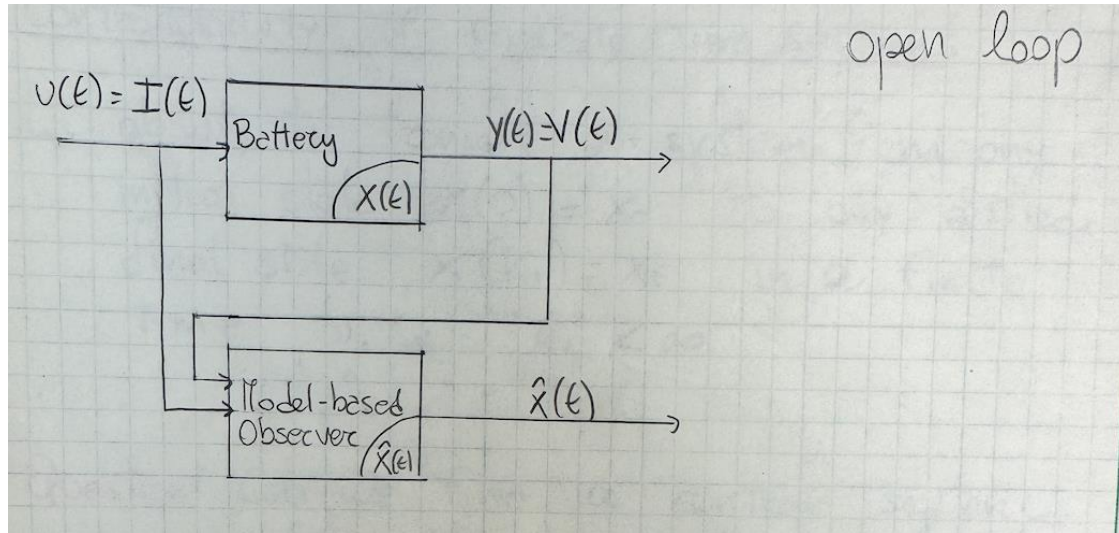
Two primary approaches

Open-loop

versus

Closed-loop

Luenberger observer



Open-loop estimator

Given:

$$\dot{x}(t) = Ax(t) + Bu(t),$$

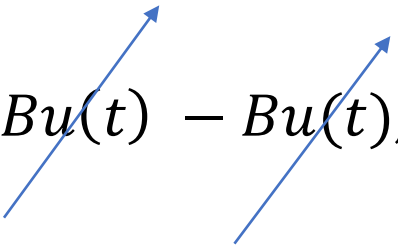
and

$$\dot{\hat{x}}(t) = A \hat{x}(t) + Bu(t),$$

- Define the estimation error as

$$\tilde{x}(t) = x(t) - \hat{x}(t)$$

Let's subtract $\dot{\hat{x}}(t)$ from $\dot{x}(t)$

$$\frac{d(x(t) - \hat{x}(t))}{dt} = A\tilde{x}(t) + Bu(t) - Bu(t),$$


Which has the solution $\tilde{x}(t) = e^{At} \tilde{x}(0)$

Open-loop estimator

$$\tilde{x}(t) = e^{At} \tilde{x}(0)$$

- If $\tilde{x}(0) = 0$, then the estimation error goes to zero
- If $\tilde{x}(0) \neq 0$, $\tilde{x}(t) \rightarrow 0$ only if A is stable (eigenvalues of A all with negative real part)
 - The dynamics of the estimation error are completely determined by the open-loop dynamics of the system
 - Could be very slow
 - There is not a way to modify the estimation error dynamics

WE SHOULD NOT USE THE OPEN-LOOP ESTIMATOR (ever)

Closed-loop estimator

- Let's use available information

$$y(t) = Cx(t) \rightarrow \text{measured output}$$

and define

$$\hat{y}(t) = C\hat{x}(t) \rightarrow \text{estimated output}$$

Feedback $\hat{y}(t)$ to improve the estimate of the state.

Closed-loop estimator

Plant	Replica
$\frac{dx}{dt} = Ax(t) + Bu(t)$ $y(t) = Cx(t)$	$\frac{d\hat{x}}{dt} = A\hat{x}(t) + Bu(t)$ $\hat{y}(t) = C\hat{x}(t)$

- Compare $y(t) = Cx(t)$ with $\hat{y}(t) = C\hat{x}(t)$
- Form $\tilde{y} = y(t) - \hat{y}(t) \equiv C(x(t) - \hat{x}(t))$

Closed-loop estimator

Plant	Replica
$\begin{aligned}\frac{dx}{dt} &= Ax(t) + Bu(t) \\ y(t) &= Cx(t)\end{aligned}$	$\begin{aligned}\frac{d\hat{x}}{dt} &= A\hat{x}(t) + Bu(t) \\ \hat{y}(t) &= C\hat{x}(t)\end{aligned}$

- Compare $y(t) = Cx(t)$ with $\hat{y}(t) = C\hat{x}(t)$
- Form $\tilde{y} = y(t) - \hat{y}(t) \equiv C(x(t) - \hat{x}(t))$

Basic form of the estimator is

$$\begin{aligned}\frac{d\hat{x}}{dt} &= A\hat{x}(t) + Bu(t) + L\tilde{y}(t) \\ \hat{y}(t) &= C\hat{x}(t)\end{aligned}$$

Where L is the observer gain (selected by the designer)

Closed-loop estimator

Let's define the error dynamics as:

$$\begin{aligned}\dot{\tilde{x}}(t) &= \dot{x}(t) - \dot{\hat{x}}(t) \\ &= [Ax(t) + Bu(t)] - [A\hat{x}(t) + Bu(t) + L(y(t) - \hat{y}(t))] \\ &= A(x(t) - \hat{x}(t)) + L(Cx(t) - C\hat{x}(t)) \\ &= A\tilde{x}(t) - LC\tilde{x}(t) \\ &= (A - LC)\tilde{x}\end{aligned}$$

So the closed-loop estimation error dynamics is now

$$\dot{\tilde{x}}(t) = (A - LC)\tilde{x}$$

with solution

$$\tilde{x}(t) = e^{(A-LC)t}\tilde{x}(0)$$

Closed-loop estimator

- We can select the gain L to attempt to improve the convergence of the estimation error (and/or speed it up) iff system is observable

$$\begin{aligned}\frac{d\hat{x}}{dt} &= A\hat{x}(t) + Bu(t) + L\tilde{y}(t) \\ &= (A - LC)\hat{x} + \underbrace{Bu(t) + Ly(t)}_{\text{Inputs to the estimator}}\end{aligned}$$
$$\hat{y} = C\hat{x}(t)$$

- The closed-loop state estimator is a dynamic system with eigenvalues given by $\lambda_i(A - LC)$ and which takes the measured plant outputs as an input and generates $\hat{x}(t)$, i.e. an estimate of $x(t)$

Example – Luenberger Observer

- Design a closed-loop observer for the system:

$$\frac{d^2x(t)}{dt^2} + \frac{dx(t)}{dt} + x(t) - u(t) = 0$$

Initial values:

$$x(0) = 0, \quad \frac{dx(0)}{dt} = 0$$

$u(t)$ is step input of magnitude 2

Value of $x(t)$ is available via sensor.

Open-loop system response

$$x_0 = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

where

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$D = 0 \quad L \in R^{2 \times 1}$$

% True initial conditions (same for all cases)

x_0 = [0; 0];

% Time vector

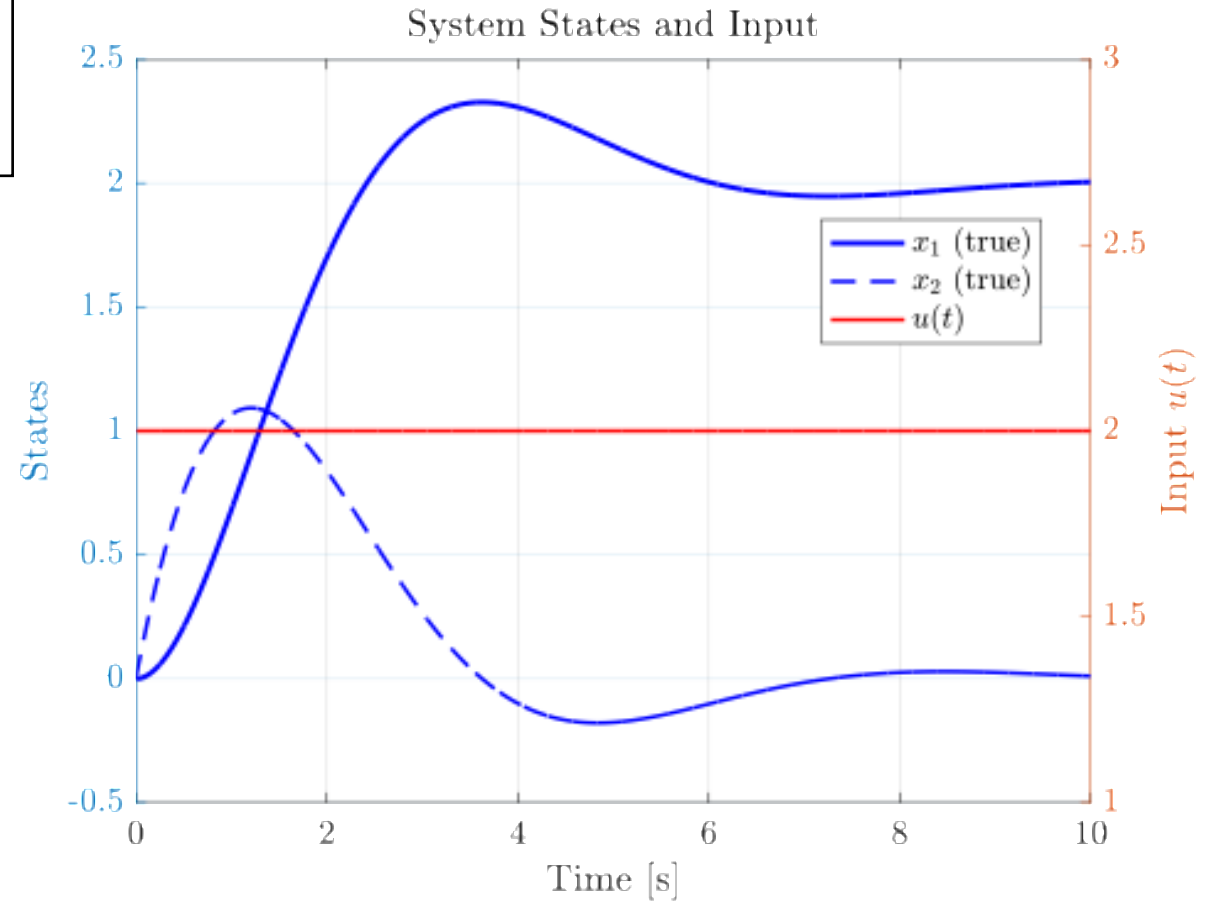
t = linspace(0, 10, 5000)';

% Input: step of magnitude 2

u = 2 * ones(size(t));

syst = ss(A, B, C, D);

[t_ss, x_ss] = lsim(syst, u, t, x_0);



$$\left. \begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned} \right\} \text{ True System}$$

$$\left. \begin{aligned} \dot{\hat{x}} &= (A - LC)\hat{x} + Bu + LCx \\ \hat{y} &= C\hat{x} \end{aligned} \right\} \text{ Observer}$$

Augmented state space system:

$$\dot{x}_{aug} = \underbrace{\begin{bmatrix} A & 0_{2 \times 2} \\ LC & A - LC \end{bmatrix}}_{A_{aug}} \underbrace{\begin{bmatrix} x \\ \hat{x} \end{bmatrix}}_{x_{aug}} + \underbrace{\begin{bmatrix} B \\ B \end{bmatrix}}_{B_{aug}} u$$

$$y_{aug} = \underbrace{\begin{bmatrix} C & 0_{1 \times 2} \\ 0_{1 \times 2} & C \end{bmatrix}}_{A_{aug}} \underbrace{\begin{bmatrix} x \\ \hat{x} \end{bmatrix}}_{x_{aug}} + \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{D_{aug}} u$$

where $x_{aug} = \begin{bmatrix} x \\ \hat{x} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$, and $y_{aug} = \begin{bmatrix} y \\ \hat{y} \end{bmatrix}$

$$x_{aug,0} = \begin{bmatrix} x_1(0) \\ x_2(0) \\ \hat{x}_1(0) \\ \hat{x}_2(0) \end{bmatrix}$$

True initial states $\rightarrow$ unknown

Estimated initial states $\rightarrow$ known

```
% Find closed loop feedback gain  
p = [-3 -3]; % desired pole/eigenvalue locations  
L_gain = place(A',C',p)'; % compute observer gain  
% eig_observer = eig(A - L_gain*C);
```

$$L \text{ (Observer gain)} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

```
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```

$$L \text{ (Observer gain)} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Simulate the above system for three cases of incorrect initial observer states.

Case 1: $x_{aug}(0) = \begin{bmatrix} 0 \\ 0 \\ -1 \\ -3 \end{bmatrix}$; **Case 2:** $x_{aug}(0) = \begin{bmatrix} 0 \\ 0 \\ -6 \\ -10 \end{bmatrix}$; **Case 3:** $x_{aug}(0) = \begin{bmatrix} 0 \\ 0 \\ -15 \\ -20 \end{bmatrix}$

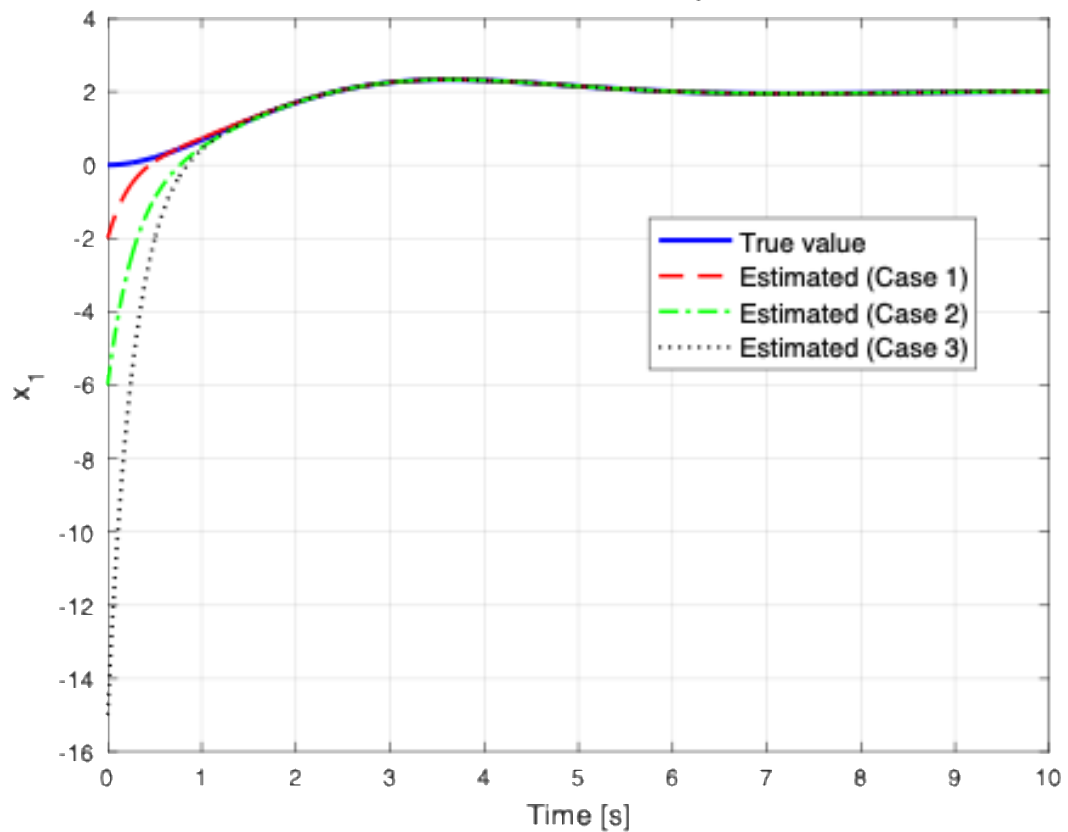
Matlab implementation:

```
xAug_0 = [x_0; xhat_0]; % Initial states of augmented system
aug_syst = ss(A_aug, B_aug, C_aug, D_aug); % define state space
[y_aug, t_out, x_aug] = lsim(aug_syst,u,t,xAug_0); % Simulate linear system
```

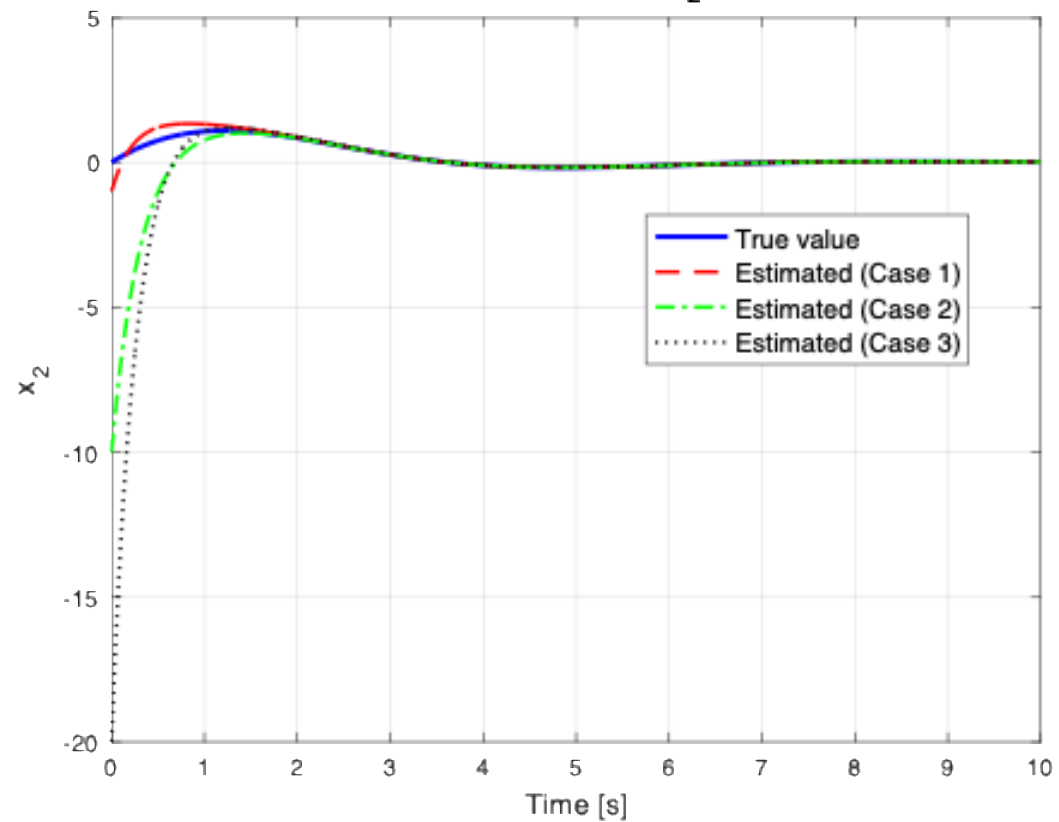


All 4 states in this variable

Estimation of x_1



Estimation of x_2



Case 1: $\hat{x}(0) = [-2, -1]$

Case 2: $\hat{x}(0) = [-6, -10]$

Case 3: $\hat{x}(0) = [-15, -20]$

Error Dynamics

Extension: Augment estimation error dynamics:

$$\left. \begin{array}{l} \dot{x} = Ax + Bu \\ y = Cx \end{array} \right\} \quad \textbf{\underline{True System}}$$

$$\left. \begin{array}{l} \dot{\hat{x}} = (A - LC)\hat{x} + Bu + LCx \\ \hat{y} = C\hat{x} \end{array} \right\} \quad \textbf{\underline{Observer}}$$

$$\dot{e} = (A - LC)e \quad \left. \right\} \quad \textbf{\underline{Error dynamics}} \quad (\text{where } e = x - \hat{x})$$

Augmented state space system:

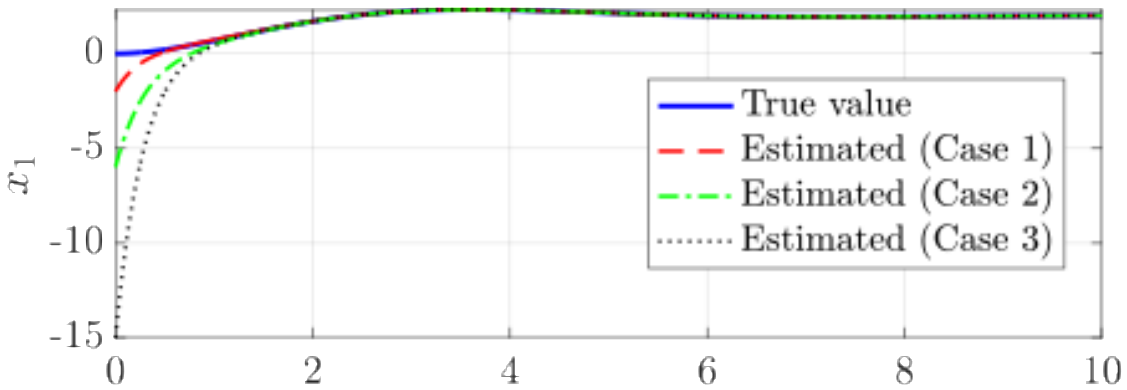
$$\dot{x}_{aug} = \begin{bmatrix} A & 0_{2 \times 2} & 0_{2 \times 2} \\ LC & A - LC & 0_{2 \times 2} \\ 0_{2 \times 2} & 0_{2 \times 2} & A - LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \\ e \end{bmatrix} + \begin{bmatrix} B \\ B \\ 0_{2 \times 1} \end{bmatrix} u$$

$$y_{aug} = \begin{bmatrix} C & 0_{1 \times 2} & 0_{1 \times 2} \\ 0_{1 \times 2} & C & 0_{1 \times 2} \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \\ e \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u$$

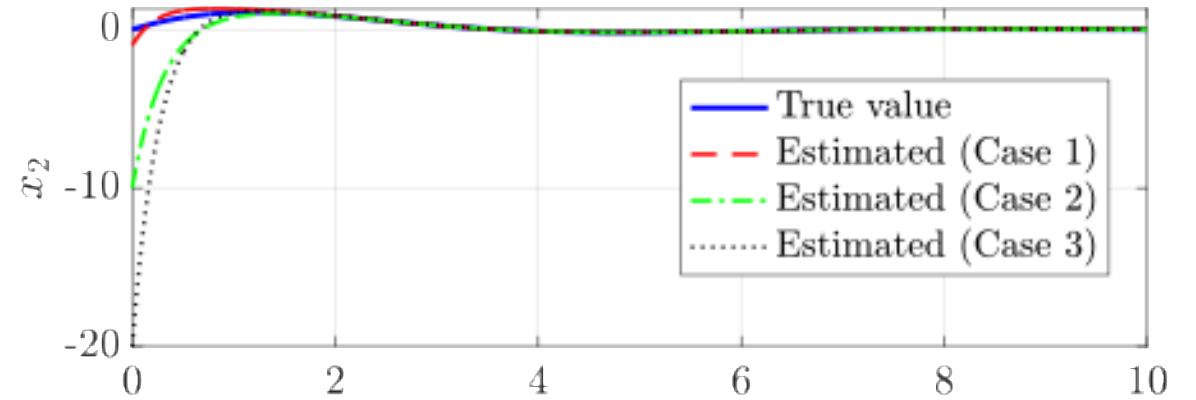
$$\text{where } x_{aug} = \begin{bmatrix} x \\ \hat{x} \\ e \end{bmatrix}, \text{ and } y_{aug} = \begin{bmatrix} y \\ \hat{y} \end{bmatrix}$$

Error Dynamics

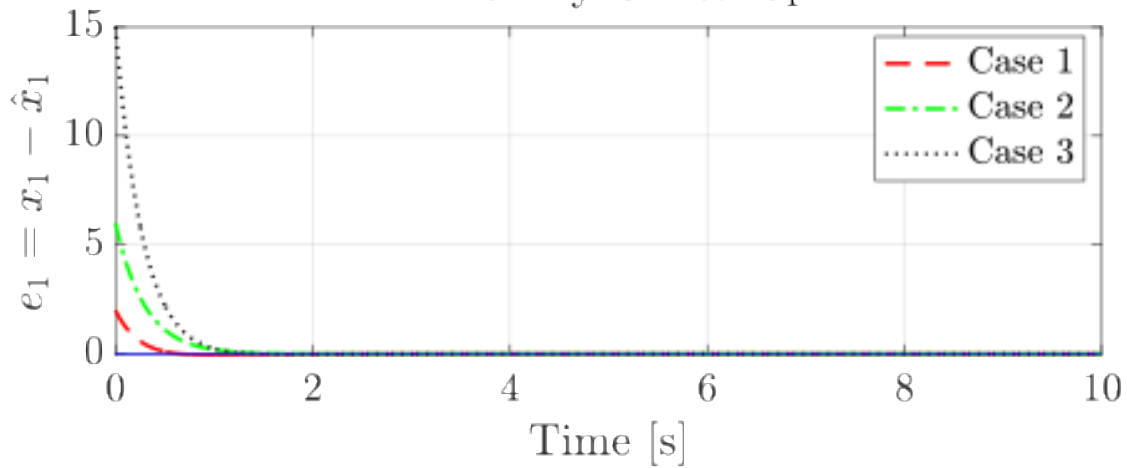
Estimation of x_1



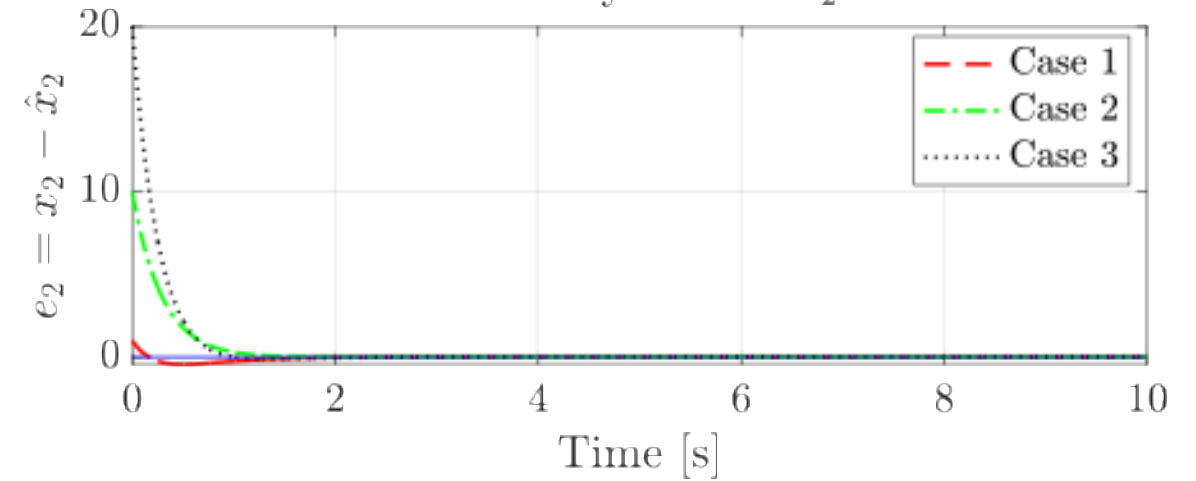
Estimation of x_2



Error Dynamics: e_1



Error Dynamics: e_2



Sensor placement

Assuming only one state can be measured, which sensor placement leads to better state reconstruction performance?

Sensor placement

Assuming only one state can be measured, which sensor placement leads to better state reconstruction performance?

Specifically, compare the two choices:

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \text{ measure } x_1($$

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix} \text{ measure } x_2($$

→ introduce the concept of **condition number** to evaluate the two cases.

Issues with Observability Matrix

Observability observability is a binary test

$$\begin{aligned}\text{rank}(\mathcal{O}) = n &\Rightarrow \text{observable} \\ \text{rank}(\mathcal{O}) < n &\Rightarrow \text{not observable}\end{aligned}$$

That's a yes/no statement.

- It answers the theoretical question: *Can I reconstruct the state?* does the mathematical structure of the system guarantee that the state is observable?

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That's a yes/no statement.

- It answers the theoretical question: *Can I reconstruct the state?* does the mathematical structure of the system guarantee that the state is observable?
- It does not answer the practical question: *Will I reconstruct the state accurately in practice?*
 - Even if a system is observable in theory, numerical issues, noise, model uncertainty, and poor conditioning can make the actual state reconstruction unreliable or highly sensitive in real implementations.

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- It answers the theoretical question: *Can I reconstruct the state?* does the mathematical structure of the system guarantee that the state is observable?
- It does not answer the practical question: *Will I reconstruct the state accurately in practice?*
 - Even if a system is observable in theory, numerical issues, noise, model uncertainty, and poor conditioning can make the actual state reconstruction unreliable or highly sensitive in real implementations.
- In practice, we need to invert a matrix, and if its determinant is close to zero, the inversion becomes numerically unstable.

Condition number

Definition

For a matrix O , the condition number (with respect to a norm) is:

$$\kappa(O) = \|O\| \cdot \|O^{-1}\|$$

If we use the 2-norm (which is most common in control and estimation), then:

$$\kappa_2(O) = \frac{\sigma_{\max}(O)}{\sigma_{\min}(O)}$$

where:

$\sigma_{\max}$ = largest singular value

$\sigma_{\min}$ = smallest singular value

Small $\kappa(O)$ $\rightarrow$ state reconstruction is numerically reliable

Large $\kappa(O)$ $\rightarrow$ theoretically observable but practically noisy

The singular values satisfy:

$$\sigma_{\max}(O) = \max_{\|x\|=1} \|Ox\|$$
$$\sigma_{\min}(O) = \min_{\|x\|=1} \|Ox\|$$

They represent:

- The **maximum amplification** of a unit state direction
- The **minimum amplification** of a unit state direction
- The ratio of these two values (i.e., the condition number) measures the range of amplification the matrix O can apply.
- *The condition number measures how much relative error in the input can be amplified in the output.*
- A high condition number means there's a big difference between the strongest and weakest amplification directions, indicating that the system can distort or amplify errors significantly in certain directions.

Condition number

The condition number measures how much relative error in the input can be amplified in the output.

If you solve:

$$Ox = y$$

and there is small noise δy , the relative error satisfies approximately:

$$\frac{\| \delta x \|}{\| x \|} \leq \kappa(O) \frac{\| \delta y \|}{\| y \|}$$

So:

$\kappa \approx 1 \rightarrow$ very stable inversion

$\kappa \gg 1 \rightarrow$ small noise gets amplified

$\kappa \rightarrow \infty \rightarrow$ matrix nearly singular

Sensor placement cont'd

Case 1: measure x_1 $C = [1 \ 0]$

$$\mathcal{O}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Rank: $\text{rank}(\mathcal{O}_1) = 2 \Rightarrow$ observable.

Condition number:

$$\kappa_1(\mathcal{O}_1) = \kappa_1(I) = 1$$

Very well-conditioned. Errors in the measurements are *not* amplified

Matlab

```
O = obsv(A,C);
```

```
k = cond(O);
```

Sensor placement cont'd

Case 2: measure x_2 $C = [0 \ 1]$

$$\mathcal{O}_2 = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

Rank: $\text{rank}(\mathcal{O}_2) = 2 \Rightarrow$ observable.

Condition number,

$$\kappa_2(\mathcal{O}_2) = \frac{\sigma_{\max}(\mathcal{O}_2)}{\sigma_{\min}(\mathcal{O}_2)},$$

Singular values come from eigenvalues of $\mathcal{O}_2^T \mathcal{O}_2$

Sensor placement cont'd

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Step 1: Compute $\mathcal{O}_2^T \mathcal{O}_2$

$$\mathcal{O}_2^T = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$$
$$\mathcal{O}_2^T \mathcal{O}_2 = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Sensor placement cont'd

Case 2: measure x_2 $C = [0 \ 1]$

$$O_2 = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

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Step 2: Eigenvalues of $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$

Solve

$$\det\left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - \lambda I\right) = 0 \Rightarrow \lambda^2 - 3\lambda + 1 = 0$$

$$\lambda = \frac{3 \pm \sqrt{5}}{2} \rightarrow \text{So } \lambda_{\max} = \frac{3 + \sqrt{5}}{2}, \lambda_{\min} = \frac{3 - \sqrt{5}}{2}$$

Step 3: Singular values

$$\sigma_{\max}(O_2) = \sqrt{\lambda_{\max}(O_2^T O_2)} = \sqrt{\frac{3 + \sqrt{5}}{2}} = 1.618$$

$$\sigma_{\min}(O_2) = \sqrt{\lambda_{\min}(O_2^T O_2)} = \sqrt{\frac{3 - \sqrt{5}}{2}} = 0.618$$

Sensor placement cont'd

Case 2: measure x_2 $C = [0 \ 1]$

$$O_2 = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

Rank: $\text{rank}(O_2) = 2 \Rightarrow$ observable.

Condition number,

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Singular values come from eigenvalues of $O_2^T O_2$

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Step 1: Compute $O_2^T O_2$

$$O_2^T = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$$
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$$\sigma_{\min}(O_2) = \sqrt{\lambda_{\min}(O_2^T O_2)} = \sqrt{\frac{3 - \sqrt{5}}{2}} = 0.618$$

Step 4: Condition number

$$\kappa_2(O_2) = \frac{1.618}{0.618} \approx 2.618 > 1$$

If your sensor noise is $\sim 1\%$ relative:

$\kappa = 1 \rightarrow$ worst-case state error $\sim 1\%$

$\kappa = 2.86 \rightarrow$ worst-case state error $\sim 2.86\%$

Important :

The condition number does not tell you how much noise will propagate in practice.
It tells you the worst-case amplification direction.

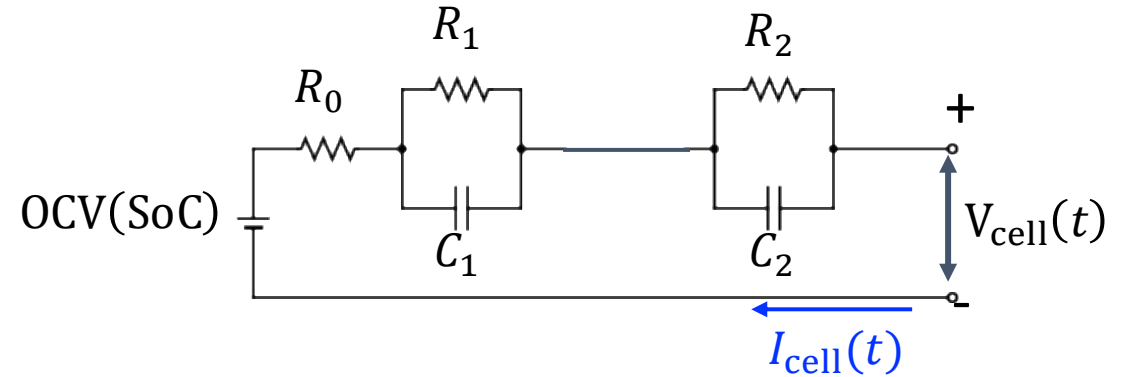
In many directions the amplification is smaller — but in the most unfavorable direction, it can reach κ .

2nd order Equivalent Circuit Model (ECM)

State equations

$$\begin{aligned}\dot{SOC} &= -\frac{1}{Q}I \\ \dot{V}_1 &= -\frac{1}{R_1 C_1} V_1 + \frac{1}{C_1} I \\ \dot{V}_2 &= -\frac{1}{R_2 C_2} V_2 + \frac{1}{C_2} I\end{aligned}$$

$$x = \begin{bmatrix} SOC \\ V_1 \\ V_2 \end{bmatrix}, u = I$$



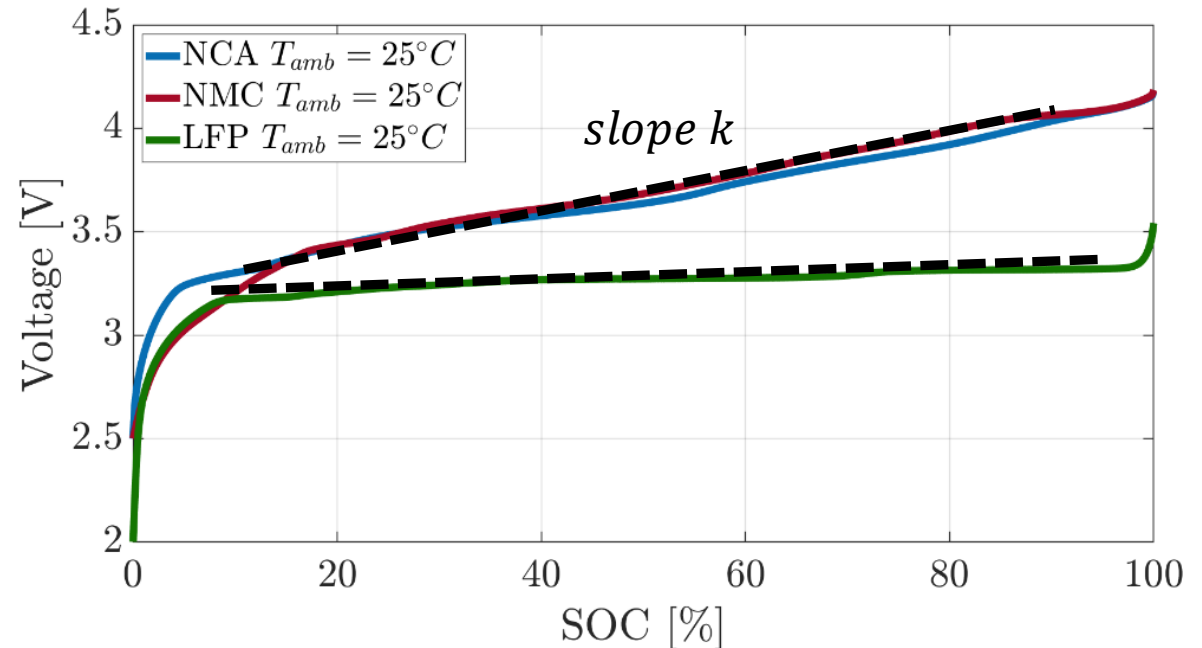
Output (KVL)

$$V = OCV(SOC) - R_0 I - V_1 - V_2$$

Let

$$k = \frac{dOCV}{dz} \big|_{soc}$$

$$OCV = k \cdot SOC$$



2nd order ECM

(linearized) State Space

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\frac{1}{R_1 C_1} & 0 \\ 0 & 0 & -\frac{1}{R_2 C_2} \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{1}{Q} \\ \frac{1}{C_1} \\ \frac{1}{C_2} \end{bmatrix}$$

$$C = [k \quad -1 \quad -1], \quad D = [-R_0]$$

2nd order ECM - Observability

For a 3-state LTI (linearized) system

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

$$CA = \begin{bmatrix} 0 & \frac{1}{R_1 C_1} & \frac{1}{R_2 C_2} \end{bmatrix},$$
$$CA^2 = \begin{bmatrix} 0 & -\left(\frac{1}{R_1 C_1}\right)^2 & -\left(\frac{1}{R_2 C_2}\right)^2 \end{bmatrix}$$

$$\mathcal{O} = \begin{bmatrix} k & \frac{-1}{R_1 C_1} & \frac{-1}{R_2 C_2} \\ 0 & \frac{1}{R_1 C_1} & \frac{1}{R_2 C_2} \\ 0 & -\left(\frac{1}{R_1 C_1}\right)^2 & -\left(\frac{1}{R_2 C_2}\right)^2 \end{bmatrix}$$

Rank of $\mathcal{O}$

The rank is **3 (full rank)** if and only if:

$$k \neq 0 \text{ and } a_1 \neq a_2$$

If $k = \frac{dOCV}{dSOC}$ is very small but not zero, then:

- $\mathcal{O}$ is full rank (rank 3) in theory
- the smallest singular value $\sigma_{min}(\mathcal{O})$ becomes very small, so $\kappa(\mathcal{O}) = \frac{\sigma_{max}}{\sigma_{min}}$ becomes very large $\rightarrow$ high noise amplification in state reconstruction
- $k \approx 0$ Voltage gives you almost no SOC information, any attempt to infer SOC from V becomes extremely noise-sensitive.
 $\Rightarrow$ SOC is observable but extremely ill-conditioned $\Rightarrow$ SOC estimate is fragile / noise-amplifying, even if V_1, V_2 are still well estimated.

